

CMSC411: Computer Systems Architecture
Project 1

Due: 03/06/2024, 11:59 pm, EST (via GradeScope)

Submit a .cpp and a .pdf file to GradeScope named "<UID>_prj1.pdf"

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bzip2.trc

Branch Predictor Settings

Predictor Type: B

Entries: 64

Bits Per Counter: 2

Bits Per History: 2

Branch Predictor Statistics

Branches: 94205

Predicted Taken: 54757

Predicted Not Taken: 39448

Correct: 90057

Misprediction Rate: 0.044032

Storage overhead: 128 bits

Branch Predictor Settings

Predictor Type: L

Entries: 64

Bits Per Counter: 2

Bits Per History: 4

Branch Predictor Statistics

Branches: 94205

Predicted Taken: 54513

Predicted Not Taken: 39692

Correct: 90511

Misprediction Rate: 0.039212

Storage overhead: 2304 bits

Branch Predictor Settings

Predictor Type: T

Entries: 64

Bits Per Counter: 2

Bits Per History: 4

Branch Predictor Statistics

Branches: 94205

Predicted Taken: 54455

Predicted Not Taken: 39750

Correct: 89905

Misprediction Rate: 0.045645

Storage overhead: 288 bits

Of the branch predictors tested in the project:

The first best is the Local history, 64 entries, 2 counter bits, 4 history bits.

This branch predictor type has the most correct number of branches predicted compared to the other types, with different entries, counter bits and history bits.

The second best predictor type is Bimodal, 64 entries, 2 counter bits, 2 history bits.

These two predictors had over 90,000 predictions correct of 94,205 total branches and had a misprediction rate of less than 5%.

The third best predictor is the Two Level Adaptive, 64 entries, 2 counter bits, 4 history bits.

It did not reach 90,000 branches correct, but it got very close with 89,905 out of 94205 branches correct, and still a misprediction rate of less than 5%

Other variations:

Branch Predictor Settings

Predictor Type: L

Entries: 256

Bits Per Counter: 3

Bits Per History: 2

Branch Predictor Statistics

Branches: 94205

Predicted Taken: 55243

Predicted Not Taken: 38962

Correct: 90997

Misprediction Rate: 0.034053

Storage overhead: 3584 bits

Branch Predictor Settings

Predictor Type: T

Entries: 512

Bits Per Counter: 4

Bits Per History: 7

Branch Predictor Statistics

Branches: 94205

Predicted Taken: 55152

Predicted Not Taken: 39053
Correct: 90592
Misprediction Rate: 0.038353
Storage overhead: 4096 bits

Branch Predictor Settings
Predictor Type: B
Entries: 1024
Bits Per Counter: 3
Bits Per History: 0

Branch Predictor Statistics
Branches: 94205
Predicted Taken: 56099
Predicted Not Taken: 38106
Correct: 90901
Misprediction Rate: 0.035072
Storage overhead: 3072 bits

Branch Predictor Settings
Predictor Type: G
Entries: 512
Bits Per Counter: 3
Bits Per History: 0

Branch Predictor Statistics
Branches: 94205
Predicted Taken: 54942
Predicted Not Taken: 39263
Correct: 90686
Misprediction Rate: 0.037355
Storage overhead: 1536 bits

These were the best ones I could find with brute force for other variations of the branch types.
The best was Local History, entries 256, counter bits 3, history bits 2.
It got 90997 predictions out of 94205 correct, and had a misprediction rate of less than %3.5

gcc.trc

Branch Predictor Settings

Predictor Type: B
Entries: 64
Bits Per Counter: 2
Bits Per History: 2

Branch Predictor Statistics
Branches: 316096
Predicted Taken: 137129
Predicted Not Taken: 178967
Correct: 291689
Misprediction Rate: 0.077214
Storage overhead: 128 bits

Branch Predictor Settings
Predictor Type: L
Entries: 64
Bits Per Counter: 2
Bits Per History: 4

Branch Predictor Statistics
Branches: 316096
Predicted Taken: 134291
Predicted Not Taken: 181805
Correct: 297567
Misprediction Rate: 0.058618
Storage overhead: 2304 bits

Branch Predictor Settings
Predictor Type: T
Entries: 64
Bits Per Counter: 2
Bits Per History: 4

Branch Predictor Statistics
Branches: 316096
Predicted Taken: 134124
Predicted Not Taken: 181972
Correct: 294408
Misprediction Rate: 0.068612
Storage overhead: 288 bits

Of the branch predictors tested in the project:
The first best is the Local history, 64 entries, 2 counter bits, 4 history bits.

This branch predictor type has the most correct number of branches predicted compared to the other types, with different entries, counter bits and history bits.

The second best predictor type is Two Level Adaptation, 64 entries, 2 counter bits, 4 history bits.

The third best predictor is the Bimodal, 64 entries, 2 counter bits, 2 history bits.

These three predictors had over 290,000 predictions correct of 316,096 total branches and had a misprediction rate of less than 8%.

Other variations:

Branch Predictor Settings

Predictor Type: T

Entries: 512

Bits Per Counter: 4

Bits Per History: 7

Branch Predictor Statistics

Branches: 316096

Predicted Taken: 138260

Predicted Not Taken: 177836

Correct: 298196

Misprediction Rate: 0.056628

Storage overhead: 4096 bits

Branch Predictor Settings

Predictor Type: G

Entries: 512

Bits Per Counter: 2

Bits Per History: 0

Branch Predictor Statistics

Branches: 316096

Predicted Taken: 135077

Predicted Not Taken: 181019

Correct: 297013

Misprediction Rate: 0.060371

Storage overhead: 1024 bits

Branch Predictor Settings

Predictor Type: B

Entries: 2048

Bits Per Counter: 2

Bits Per History: 0

Branch Predictor Statistics

Branches: 316096

Predicted Taken: 135234
Predicted Not Taken: 180862
Correct: 297660
Misprediction Rate: 0.058324
Storage overhead: 4096 bits

Branch Predictor Settings
Predictor Type: L
Entries: 256
Bits Per Counter: 3
Bits Per History: 2

Branch Predictor Statistics
Branches: 316096
Predicted Taken: 136494
Predicted Not Taken: 179602
Correct: 299452
Misprediction Rate: 0.052655
Storage overhead: 3584 bits

These are the best ones I could find with brute force for other variations of the branch types.
The best was Local History, entries 256, counter bits 3, history bits 2.
It got 299452 predictions out of 316096 correct, and had a misprediction rate of about %5.

h264ref.trc

Branch Predictor Settings
Predictor Type: L
Entries: 64
Bits Per Counter: 1
Bits Per History: 4

Branch Predictor Statistics
Branches: 220586
Predicted Taken: 133385
Predicted Not Taken: 87201
Correct: 203964
Misprediction Rate: 0.075354
Storage overhead: 1280 bits

Branch Predictor Settings

Predictor Type: L

Entries: 64

Bits Per Counter: 2

Bits Per History: 4

Branch Predictor Statistics

Branches: 220586

Predicted Taken: 134374

Predicted Not Taken: 86212

Correct: 207917

Misprediction Rate: 0.057433

Storage overhead: 2304 bits

Branch Predictor Settings

Predictor Type: T

Entries: 64

Bits Per Counter: 2

Bits Per History: 4

Branch Predictor Statistics

Branches: 220586

Predicted Taken: 134183

Predicted Not Taken: 86403

Correct: 199878

Misprediction Rate: 0.093877

Storage overhead: 288 bits

Of the branch predictors tested in the project:

The first best is the Local history, 64 entries, 2 counter bits, 4 history bits.

This branch predictor type has the most correct number of branches predicted compared to the other types, with different entries, counter bits and history bits.

The second best predictor type is Local history, 64 entries, 1 counter bits, 4 history bits.

The first two best had branch predictions over 200,000 out of 220586 branches, and the first had a misprediction rate of about %5, while the second had a %7

The third best predictor is the Two Level Adaptive, 64 entries, 2 counter bits, 2 history bits.

This prediction had over 199,000 predictions correct and had a misprediction rate of less than 10%.

Other variations:

Branch Predictor Settings

Predictor Type: B

Entries: 2048

Bits Per Counter: 2

Bits Per History: 0

Branch Predictor Statistics

Branches: 220586

Predicted Taken: 132230

Predicted Not Taken: 88356

Correct: 209327

Misprediction Rate: 0.051041

Storage overhead: 4096 bits

Branch Predictor Settings

Predictor Type: G

Entries: 2048

Bits Per Counter: 2

Bits Per History: 0

Branch Predictor Statistics

Branches: 220586

Predicted Taken: 134051

Predicted Not Taken: 86535

Correct: 210522

Misprediction Rate: 0.045624

Storage overhead: 4096 bits

Branch Predictor Settings

Predictor Type: T

Entries: 512

Bits Per Counter: 3

Bits Per History: 7

Branch Predictor Statistics

Branches: 220586

Predicted Taken: 135253

Predicted Not Taken: 85333

Correct: 210636

Misprediction Rate: 0.045107

Storage overhead: 3968 bits

Branch Predictor Settings

Predictor Type: L

Entries: 128

Bits Per Counter: 3

Bits Per History: 3

Branch Predictor Statistics
Branches: 220586
Predicted Taken: 135247
Predicted Not Taken: 85339
Correct: 209450
Misprediction Rate: 0.050484
Storage overhead: 3456 bits

These are the best ones I could find with brute force for other variations of the branch types.
The best was Two Level Adaptive, entries 512, counter bits 3, history bits 7.
It got 210636 predictions out of 220586 correct, and had a misprediction rate of about %4.5.